

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5: *Measure network performance using key metrics with simulation tools*

Assessment Tool Weightage: 20%

QUESTIONS: 40

Total Marks:40

Annexure A

Mock Interview

Code of Conduct:

*I **C.BHUVANESH (192521108)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: c.bhuvanesh

QUESTIONS:40

Total Marks:40

1. Analyze how network congestion affects packet delay and packet loss. (BL4)(1)
2. Compare the impact of star and mesh topologies on network reliability and fault tolerance. (BL4)(1)
3. Why does increasing the number of routers between source and destination increase end-to-end delay? Explain logically. (BL4)(1)
4. Analyze how Quality of Service (QoS) improves the performance of voice and video traffic. (BL4)(1)

5. Evaluate the impact of using fiber optic cables instead of copper cables for high-speed communication. (BL5)(1)
6. Analyze how network segmentation using VLANs improves performance and security. (BL4)(1)
7. Compare the effects of TCP and UDP on network performance for different applications. (BL4)(1)
8. Why is load balancing important in large enterprise networks? Analyze its impact on performance and availability. (BL5)(1)
9. Analyze how frequent routing updates influence network efficiency and bandwidth utilization. (BL4)(1)
10. Evaluate the trade-off between network security measures (such as firewalls and encryption) and overall network performance. (BL5)(1)
11. A network has high delay—how will you identify and fix the issue? (BL4)(1)
12. If throughput is low despite high bandwidth, what could be the reasons? (BL4)(1)
13. How would you reduce packet loss in a congested network? (BL4)(1)
14. A VoIP call is breaking frequently—how would you troubleshoot? (BL4)(1)
15. Given a network with high jitter, suggest improvements (BL4)(1)
16. How will you simulate a congested network in Packet Tracer? (BL5)(1)
17. If packets are delayed in a router, what steps will you take? (BL4)(1)
18. How would you optimize a network for online gaming? (BL4)(1)
19. Design a small network and explain how you will measure its performance (BL5)(1)
20. If RTT is very high, how will you diagnose the issue? (BL4)(1)
21. Explain network performance to a non-technical person (BL5)(1)
22. Describe throughput using a real-life example (BL5)(1)
23. Present how packet loss affects video streaming (BL5)(1)
24. Explain delay types clearly in under 1 minute (BL4)(1)
25. Describe Cisco Packet Tracer in a professional way (BL5)(1)
26. Explain QoS in simple and technical terms (BL5)(1)
27. How would you present your simulation results to a client? (BL4)(1)
28. Explain jitter using a real-world analogy (BL5)(1)
29. Describe how you tested network performance in a lab (BL4)(1)
30. Explain the importance of performance metrics in one structured answer (BL5)(1)
31. Introduce yourself and explain your knowledge of network performance (BL5)(1)
32. Why are you interested in networking and simulation tools? (BL5)(1)
33. What makes you confident in analyzing network performance? (BL5)(1)
34. Explain a concept you are very strong in (related to networking) (BL5)(1)
35. How would you handle a question you don't know? (BL5)(1)
36. Describe a situation where you solved a technical problem (BL4)(1)
37. Why should we select you for a networking role? (BL5)(1)
38. What challenges did you face while learning Packet Tracer? (BL5)(1)
39. How do you improve your technical skills regularly? (BL5)(1)
40. Demonstrate confidence while explaining a complex concept simply (BL5)(1)

1. Analyze how network congestion affects packet delay and packet loss.

Network congestion occurs when traffic exceeds the available network capacity. Packets wait in router queues, increasing **packet delay**. When queues become full, routers discard packets, causing **packet loss**. Therefore, congestion results in higher delay, retransmissions, lower throughput, and poor application performance.

2. Compare star and mesh topologies on reliability and fault tolerance.

Star topology depends heavily on the central switch; if the switch fails, the entire network is affected. However, failure of one individual cable usually affects only one device. **Mesh topology** provides multiple paths between devices, so if one link fails, traffic can use another path. Therefore, mesh provides **higher reliability and fault tolerance**, but it is more expensive and complex.

3. Why does increasing routers increase end-to-end delay?

Every router introduces **processing delay, queuing delay, and transmission delay**. As the number of routers increases, these delays accumulate. Packets may also experience congestion at intermediate routers. Therefore, more hops generally result in higher end-to-end delay.

4. How does QoS improve voice and video traffic?

QoS prioritizes delay-sensitive traffic such as voice and video over less critical traffic. It can provide **priority queuing, bandwidth reservation, and congestion management**. This reduces delay, jitter, and packet loss, resulting in clearer voice calls and smoother video.

5. Impact of fiber optic instead of copper for high-speed communication.

Fiber provides **higher bandwidth, longer transmission distances, lower signal loss, and immunity to electromagnetic interference**. It is suitable for high-speed backbone networks. However, fiber installation and equipment can initially cost more and require specialized skills.

6. How does VLAN segmentation improve performance and security?

VLANs divide one physical network into separate logical networks. This reduces unnecessary **broadcast traffic**, improving performance. VLANs also isolate departments or users, so unauthorized users cannot directly access other segments. Therefore, VLANs improve both **network efficiency and security**.

7. Compare TCP and UDP effects on network performance.

TCP provides reliability through acknowledgments, sequencing, and retransmissions, but this introduces overhead and can increase delay. **UDP** has lower overhead and is faster but does not guarantee delivery. TCP is suitable for web browsing and file transfers, while UDP is commonly preferred for real-time applications such as gaming, VoIP, and live streaming.

8. Why is load balancing important in enterprise networks?

Load balancing distributes traffic across multiple servers or network links. It prevents one resource from becoming overloaded while others remain underused. This improves **performance, response time, scalability, and availability**. If one server or link fails, traffic can be redirected to another.

9. How do frequent routing updates influence network efficiency?

Routing updates help routers maintain accurate paths, but excessive updates consume **bandwidth, CPU, and memory**. Frequent unnecessary updates can increase network overhead. Efficient routing protocols use mechanisms such as triggered updates and route summarization to maintain current routing information without excessive overhead.

10. Trade-off between security and network performance.

Firewalls, encryption, IDS/IPS, and deep packet inspection improve security but require processing resources. This can introduce additional latency and reduce throughput if devices are overloaded. The solution is to use **appropriately sized security devices, efficient configurations, and selective inspection** to achieve a balance between security and performance.

11. A network has high delay—how will you identify and fix it?

First, use **ping and traceroute** to identify where delay occurs. Check bandwidth utilization, router CPU usage, interface errors, and queue lengths. Then identify whether the problem is congestion, faulty equipment, long routes, or insufficient bandwidth. Fix it by optimizing routing, increasing bandwidth, applying QoS, or replacing faulty equipment.

12. If throughput is low despite high bandwidth, what could be the reasons?

Possible causes include:

- Network congestion
- Packet loss and retransmissions
- High latency
- TCP limitations
- Faulty cables/interfaces

- Router or switch overload
- Server limitations
- Protocol overhead

Therefore, high theoretical bandwidth does not guarantee high actual throughput.

13. How would you reduce packet loss in a congested network?

I would first identify the congested links using monitoring tools. Then I would apply **QoS, traffic shaping, load balancing, and routing optimization**. Increasing link bandwidth or adding backup links can also help. Unnecessary traffic should be controlled to prevent congestion.

14. VoIP call is breaking frequently—how would you troubleshoot?

I would check:

1. Packet loss
2. Jitter
3. Delay
4. Available bandwidth
5. QoS configuration
6. Network congestion

I would use ping, traceroute, and packet-capture tools to identify the problem. Then prioritize VoIP traffic using QoS and fix congestion or faulty network equipment.

15. Network has high jitter—suggest improvements.

High jitter means packets are arriving at inconsistent times. I would:

- Apply QoS priority to real-time traffic.
- Reduce network congestion.
- Use traffic shaping.
- Increase bandwidth if required.
- Optimize routing.
- Use suitable jitter buffers for VoIP/video applications.

16. How will you simulate congestion in Packet Tracer?

Create several PCs communicating through routers/switches and generate heavy traffic between them. Reduce the available link bandwidth or create multiple traffic flows through the same link. Observe **queueing, delay, packet loss, and throughput** using Packet Tracer simulation mode.

17. If packets are delayed in a router, what steps will you take?

I would check:

- Router CPU and memory utilization
- Interface bandwidth
- Queue length
- Packet drops
- Interface errors
- Routing table
- QoS configuration

Then I would remove bottlenecks, optimize routing, apply QoS, or increase link capacity.

18. How would you optimize a network for online gaming?

Gaming requires **low latency, low jitter, and low packet loss**. I would prioritize gaming traffic using QoS, use efficient routing, reduce congestion, avoid overloaded Wi-Fi channels, and use wired Ethernet where possible. High-bandwidth downloads should receive lower priority during gaming.

19. Design a small network and explain how you will measure performance.

I would create:

PC1 — Switch — Router — Router — Switch — PC2

I would measure:

- **Throughput** using data transferred/time
- **Delay** using ping
- **Packet loss** using ping statistics
- **RTT** using ping
- **Jitter** by measuring variation in packet arrival times

I would compare the results against the required performance levels.

20. If RTT is very high, how will you diagnose it?

I would use **ping** to measure RTT and **tracert** to identify the hop causing the delay. Then I would check congestion, routing paths, link utilization, interface errors, and device CPU usage. If necessary, I would optimize the route or increase network capacity.

21. Explain network performance to a non-technical person.

Network performance describes **how well a network delivers information**. It is similar to a road system: bandwidth is like the number of lanes, delay is the travel time, packet loss is like

vehicles losing their deliveries, and throughput is the amount of information successfully delivered.

22. Describe throughput using a real-life example.

Throughput is the amount of data successfully transferred in a given time. For example, if a network transfers **8000 bits in 4 seconds**:

Throughput = $\frac{8000 \text{ bits}}{4 \text{ seconds}} = 2000 \text{ bps}$

It is similar to measuring how many vehicles successfully pass through a road every second.

23. How does packet loss affect video streaming?

Packet loss causes parts of the video data to disappear. This can produce **freezing, buffering, pixelation, missing audio, and reduced video quality**. A small amount may be corrected, but high packet loss significantly affects streaming quality.

24. Explain delay types clearly in under 1 minute.

There are four major types of network delay:

- **Processing delay:** Time taken by a device to process a packet.
- **Queuing delay:** Time the packet waits in a queue.
- **Transmission delay:** Time required to place the packet onto the link.
- **Propagation delay:** Time required for the signal to travel through the medium.

Total delay is the combination of these delays.

25. Describe Cisco Packet Tracer professionally.

Cisco Packet Tracer is a network simulation and learning tool used to design, configure, and troubleshoot virtual network environments. It allows users to work with routers, switches, PCs, protocols, and network services without requiring physical hardware.

26. Explain QoS in simple and technical terms.

Simple: QoS is like giving an emergency vehicle priority on a busy road.

Technical: Quality of Service classifies and prioritizes network traffic to provide appropriate **bandwidth, delay, jitter, and packet-loss characteristics**. It is especially useful for voice, video, and other real-time applications.

27. How would you present simulation results to a client?

I would present:

1. Network design
2. Test methodology
3. Performance metrics
4. Results using graphs/tables
5. Problems identified
6. Recommended improvements
7. Before-and-after comparison

I would focus on business impact rather than only technical terminology.

28. Explain jitter using a real-world analogy.

Imagine a bus that should arrive every **5 minutes**. If it arrives after 5 minutes, then 10 minutes, then 3 minutes, the arrival time is inconsistent. This is similar to **network jitter**, where packets that should arrive at regular intervals arrive with varying delays.

29. Describe how you tested network performance in a lab.

I created a network topology in Packet Tracer, configured IP addresses and routing, and generated traffic between devices. I measured **throughput, delay, packet loss, and RTT** using available testing and simulation tools. I then analyzed the results and identified possible bottlenecks.

30. Explain the importance of performance metrics in one structured answer.

Network performance metrics help determine whether a network is operating effectively.

- **Bandwidth:** Maximum capacity.
- **Throughput:** Actual successful data transfer.
- **Delay:** Time required for data to travel.
- **Packet loss:** Percentage of packets that do not reach the destination.
- **Jitter:** Variation in packet delay.

Together, these metrics help identify bottlenecks, troubleshoot problems, and improve network quality.

HR / Viva Questions

31. Introduce yourself and explain your knowledge of network performance.

Sample answer:

Good morning. My name is Muthu. I am interested in computer networking and network performance analysis. I have learned concepts such as bandwidth, throughput, delay, packet loss, jitter, QoS, routing, and network congestion. I have also practiced network design and simulation using Cisco Packet Tracer. I am interested in improving my practical networking skills and applying my knowledge to real-world network problems.

32. Why are you interested in networking and simulation tools?

I am interested in networking because it explains how computers and systems communicate with each other. Simulation tools such as Cisco Packet Tracer allow me to **design and test networks safely without physical hardware**. They also help me understand routing, troubleshooting, and performance concepts practically.

33. What makes you confident in analyzing network performance?

I understand the major performance metrics such as **throughput, bandwidth, delay, packet loss, jitter, and RTT**. I can analyze these measurements to identify problems such as congestion, routing issues, or insufficient capacity. I also use simulation and troubleshooting methods to verify my analysis.

34. Explain a concept you are very strong in.

One concept I am comfortable with is **network congestion**. Congestion occurs when traffic exceeds the capacity of a network link or device. It causes packets to wait in queues, increasing delay, and eventually packets may be dropped. QoS, traffic management, load balancing, and increasing bandwidth can help control congestion.

35. How would you handle a question you don't know?

I would honestly say that I don't know the answer rather than guessing. I would explain what I know about the related concept and then investigate the correct answer using reliable technical documentation or other resources. I see unfamiliar questions as opportunities to learn.

36. Describe a situation where you solved a technical problem.

Sample answer:

During a network simulation, I noticed that communication between two devices was not working. I checked the IP addresses, subnet masks, default gateways, and routing configuration step by step. I identified the configuration error, corrected it, and tested the connection again using ping. This taught me the importance of systematic troubleshooting rather than changing multiple settings at once.

37. Why should we select you for a networking role?

I have a strong interest in networking and a willingness to learn. I understand fundamental concepts such as routing, switching, QoS, congestion, and network performance metrics. I am also comfortable using simulation tools and approaching technical problems systematically. I believe my learning attitude and problem-solving skills would help me contribute and grow in the role.

38. What challenges did you face while learning Packet Tracer?

Initially, understanding router configuration, IP addressing, routing tables, and troubleshooting was challenging. I overcame this by practicing small topologies, testing configurations step by step, and analyzing errors carefully. With regular practice, I became more comfortable designing and troubleshooting networks.

39. How do you improve your technical skills regularly?

I improve my skills through **regular practice, simulations, networking exercises, documentation, and troubleshooting scenarios**. I also revise fundamental concepts and try to apply theoretical knowledge to practical network designs.

40. Demonstrate confidence while explaining a complex concept simply.

Example — TCP three-way handshake:

TCP uses a three-way handshake to establish a reliable connection. First, the client sends a SYN message to request a connection. The server responds with SYN-ACK to accept the request and synchronize communication. Finally, the client sends an ACK to confirm the response. After these three steps, both devices can reliably exchange data.

Quick revision tip: For the technical questions, remember the main performance terms as **Bandwidth → Throughput → Delay → Packet Loss → Jitter → RTT → QoS**. These seven concepts cover a large part of network-performance viva questions.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect

Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions